

Customer Promotion Response Classification Report

Executive Summary

This project develops a machine learning framework to predict whether customers will respond to promotional offers using historical retail transaction data and promotion metadata. The objective is to improve campaign targeting efficiency, reduce unnecessary promotional expenditure, and maximize marketing return on investment.

The dataset contains 20,000 historical promotion events and 34.2 million customer transaction records. After extensive leakage-free feature engineering, 71 behavioural, temporal, and contextual variables were constructed to capture customer purchasing patterns, brand affinity, store engagement, and promotional sensitivity.

Three classification models were evaluated:

Model	Validation AUC	Notes
Logistic Regression	0.7049	Linear baseline
Random Forest (200 trees)	0.7075	Non-linear ensemble baseline
LightGBM	0.7134	Selected final model

The LightGBM model achieved the strongest performance, with robust validation stability confirmed through 5-fold cross-validation (0.7063 ± 0.0111 AUC).

Business evaluation demonstrated substantial practical value:

- The top 20% of ranked customers captured 41.0% of all responders.
- This corresponds to a 2.05× lift over random outreach.
- At the recommended operating threshold, targeting precision improved from 20.2% (random targeting) to 33.1%.

Feature importance analysis showed that promotional responsiveness is driven primarily by:

- Historical shopping regularity
- Spending intensity and consistency
- Brand and category affinity
- Price sensitivity relative to customer spending patterns
- Store-level engagement behaviour

The final model demonstrates that meaningful predictive signal can be extracted from transactional retail data to support more efficient and data-driven marketing campaigns.

1. Introduction

The objective of this project is to predict whether a customer will respond to a promotional offer by leveraging historical transaction records and promotion metadata. This constitutes a binary classification problem in which accurate prediction of customer response enables marketing teams to optimize campaign targeting, reduce promotional expenditure, and maximize overall return on investment.

The dataset consists of four primary components:

Dataset	Description
train_history	Historical promotion events containing customer identifiers, promotion identifiers, store and region information, promotion dates, and the binary response variable active
test_history	Promotion events with the same structure as the training set but without response labels
transactions	Historical customer purchase records containing product category, brand, quantity, expenditure amount, and transaction date
promos	Promotion metadata containing category, manufacturer, brand, promotion quantity, and promotion value

The modelling workflow consists of three major stages:

1. Leakage-free feature engineering
2. Model selection and hyperparameter optimization
3. Feature evaluation and business interpretation

The overall objective is not only to maximize predictive performance, but also to ensure that the resulting model produces interpretable and operationally useful insights for campaign targeting.

2. Data Quality Assessment

The training dataset contains 20,000 promotion events from March 2013. Among these observations, 4,050 customers responded to promotions, corresponding to a response rate of 20.2%.

Metric	Value
Total promotion events	20,000
Positive responses	4,050
Response rate	20.2%
Negative-to-positive ratio	Approximately 3.9:1

Because of the moderate class imbalance, stratified splitting was applied throughout model development to ensure proportional representation of both classes in all training and validation partitions. No missing values were identified in either the training history or promotion metadata tables. The transaction dataset spans 34.2 million records across 29,988 unique customers between 2012-03-02 and 2013-02-28.

Dataset Characteristic	Value
Transaction records	34.2 million
Unique customers	29,988
Observation period	2012-03-02 to 2013-02-28
Missingness	2.8% in measure field only

The only missing values occur in the measure column of the transactions table. Since this variable was not used in downstream feature engineering, no imputation was required. Overall, the dataset exhibited strong structural quality and required minimal preprocessing beyond temporal filtering and feature construction.

3. Methodology

The primary modelling objective is to predict customer response to retail promotions using temporally valid behavioural signals extracted from historical transaction data.

The methodological framework consists of three stages:

1. Feature engineering
2. Model selection and hyperparameter tuning
3. Feature evaluation and selection

The modelling pipeline was designed under strict leakage-prevention principles to ensure that only information available prior to each promotion date contributed to prediction.

3.1 Feature Engineering

Feature engineering focused on constructing a broad set of leakage-free variables capturing

customer purchasing behaviour, promotional context, and store-level dynamics.

The pipeline incorporates multiple behavioural dimensions:

- Temporal shopping patterns
- Recency, frequency, and monetary (RFM) behaviour
- Brand and category affinity
- Store-level engagement
- Price sensitivity
- Shopping diversity metrics
- Competitive promotion exposure
- Customer behavioural consistency

All customer transaction histories were filtered to include only purchases occurring before the promotion date associated with each observation. This temporal filtering prevents information leakage and preserves realistic deployment conditions.

High-cardinality identifiers — including customer ID, store, product category, brand, manufacturer, and promotion ID — were encoded using Bayesian-smoothed target encoding. This approach preserves predictive signal while mitigating instability associated with low-frequency categories.

Additional interaction and normalization features were introduced to contextualize customer behaviour relative to broader store and category baselines. These include:

- Ratio-based features
- Log-transformed monetary variables
- Loyalty interaction features
- Diversity metrics
- Relative promotional value indicators

The final engineered dataset contains 71 predictive variables.

3.2 Model Selection

Three classification algorithms were evaluated on the same 80/20 stratified validation split to ensure that model selection was empirically justified.

Model	Validation AUC	Description
Logistic Regression	0.7049	Linear baseline
Random Forest (200 trees)	0.7075	Non-linear ensemble baseline
LightGBM	0.7134	Gradient boosting framework

LightGBM achieved the strongest validation performance, outperforming:

- Logistic Regression by +0.0085 AUC points

- Random Forest by +0.0059 AUC points

LightGBM's performance advantage primarily stems from its ability to:

- Capture non-linear feature interactions
- Efficiently model heterogeneous behavioural patterns
- Handle target-encoded high-cardinality variables effectively
- Operate without explicit feature scaling

Consequently, all subsequent analysis is based on the tuned LightGBM model.

Hyperparameter optimization was conducted using grid search over:

- Number of leaves
- Maximum tree depth
- Minimum child samples
- L1 regularization
- L2 regularization
- Feature subsampling ratios

Models were evaluated using validation AUC on the stratified validation set.

Early stopping with a patience of 50 rounds was applied to improve training efficiency and reduce overfitting risk.

Neural networks were not pursued because:

- The dataset is tabular rather than sequential or unstructured
- The sample size is modest relative to deep learning complexity
- Gradient boosting models are generally better suited for structured retail tabular data
- Existing results already exhibited signs of overfitting at higher model complexity

3.3 Feature Selection

Following model selection, feature importance analysis was conducted using split-based importance scores derived from the tuned LightGBM model.

Variables with consistently low importance were evaluated for removal in order to:

- Reduce dimensionality
- Minimize noise
- Improve generalization performance
- Narrow the training-validation performance gap

However, removing low-importance features did not improve validation AUC even after threshold tuning and repeated experimentation. As a result, all 71 engineered variables were retained in the final model.

The validation AUC of 0.7134 was therefore treated as the primary optimization criterion despite the moderately wider training-validation gap.

4. Feature Engineering Details

The feature engineering pipeline produces a comprehensive set of temporally valid and behaviourally informed variables that capture multiple dimensions of customer purchasing behaviour and promotional context.

4.1 Temporal Features

Temporal descriptors derived from the promotion date include:

- Day of week
- Month
- Weekend indicator

These variables capture weekly shopping patterns and seasonal behavioural effects that may influence promotional responsiveness.

4.2 Encoded Categorical Features

Several high-cardinality identifiers were encoded using Bayesian-smoothed target encoding:

- Customer ID
- Store
- Brand
- Product category
- Manufacturer
- Promotion ID

Bayesian smoothing reduces instability in low-frequency categories while preserving predictive signal from historically responsive entities.

4.3 RFM Behavioural Features

A central component of the pipeline is the generation of historical transaction aggregates using past-only customer purchases.

Recency Features

Recency variables measure the number of days since the customer's most recent transaction at several levels:

- Overall customer activity
- Brand-level activity
- Category-level activity
- Manufacturer-level activity

Frequency Features

Frequency metrics quantify customer engagement intensity, including:

- Total transaction count
- Recent 30-day transaction count
- Unique categories purchased
- Unique brands purchased

Monetary Features

Monetary variables summarize customer spending behaviour:

- Total amount spent
- Average transaction amount
- Quantity purchased
- Spending variability
- Coefficient of variation in transaction amounts

4.4 Diversity and Loyalty Metrics

Diversity metrics quantify the breadth of customer shopping behaviour across brands and categories.

Additional loyalty indicators capture:

- Brand affinity
- Category loyalty
- Share of brand purchases within category activity
- Historical engagement with promoted entities

4.5 Store-Level Features

Store-level aggregates contextualize customer behaviour relative to broader store performance.

These include:

- Average transaction amount
- Average transaction quantity
- Store transaction volume
- Historical store response rate

4.6 Price Sensitivity Features

Price sensitivity features compare promotion characteristics against customer historical spending patterns.

Examples include:

- Promotion value relative to average transaction amount
- Promotion quantity relative to historical purchase quantity
- Relative promotional attractiveness indicators

4.7 Interaction and Normalization Features

Additional interaction features combine complementary behavioural signals, including:

- Recency-weighted loyalty metrics
- Behaviour-to-store baseline ratios
- Customer spending normalization variables

Log transformations were applied to highly skewed monetary and count variables to improve separability and reduce variance.

4.8 Competitive Context Features

Short-term competitive pressure was incorporated by counting overlapping promotions within the same:

- Store
- Product category
- Recent temporal window

These variables capture promotional congestion effects that may reduce the effectiveness of individual campaigns.

5. Results

5.1 Model Performance

The final LightGBM classifier was trained using an 80/20 stratified split of the 20,000-record training dataset.

Early stopping identified an optimal convergence point at iteration 179.

Metric	Value
Training AUC	0.8751
Validation AUC	0.7134
Best iteration	179
Total features	71

The training-validation gap of approximately 0.16 AUC points reflects the increased flexibility associated with retaining the full feature set.

Although this gap indicates moderate overfitting risk, no reduced-feature configuration improved validation performance. Consequently, the full-feature model was retained, with validation AUC treated as the primary optimization objective.

Model Comparison

Model	Validation AUC	Notes
Logistic Regression	0.7049	Linear baseline
Random Forest (200 trees)	0.7075	Non-linear ensemble
LightGBM	0.7134	Selected model

5.2 Validation Stability

A single train-validation split can occasionally produce optimistic or pessimistic estimates depending on the random partition.

To confirm robustness, 5-fold stratified cross-validation was applied to the training set using:

- Identical LightGBM hyperparameters
- Fixed iteration count equal to the optimal single-split iteration
- Stratified fold construction

Cross-Validation Results

Fold	Validation AUC
Fold 1	0.7046
Fold 2	0.6902
Fold 3	0.7021
Fold 4	0.7103
Fold 5	0.7241

The mean cross-validated AUC is 0.7063 with standard deviation 0.0111.

The original single-split validation AUC of 0.7134 falls within two standard deviations of the cross-validation mean, indicating that the reported performance is not an artefact of a particularly favourable data partition.

5.3 Decision Threshold and Campaign Targeting

ROC AUC evaluates ranking performance across all possible thresholds, but operational deployment requires selection of a specific decision threshold.

A Precision-Recall analysis was therefore conducted to identify the threshold that best balances:

- Campaign reach
- Targeting precision
- Customer coverage
- Promotional efficiency

Maximization of the F1 score yielded an optimal operating threshold of 0.21.

Recommended Operating Threshold

Metric	Value
Decision threshold	0.21
F1 score	0.430
Precision	33.1%
Recall	61.3%
Customers targeted	1,199
Validation set size	4,000

At this threshold:

- 1,199 customers (37.5% of the validation set) are flagged for outreach
- 397 of 648 actual responders are correctly identified
- Targeting precision improves from 20.2% (random outreach) to 33.1%

This corresponds to a 1.64× improvement in targeting efficiency relative to random customer selection.

Marketing teams may further adjust the threshold according to campaign economics:

- Higher thresholds increase precision but reduce recall
- Lower thresholds broaden reach but reduce targeting efficiency

5.4 Business Impact – Lift and Gain

To evaluate practical business value, customers were ranked by predicted response probability and grouped into score deciles.

Decile Lift Analysis

Score Decile	Score Threshold	Responders Captured	Lift vs Random
Top 10%	0.385	25.2%	2.52×
Top 20%	0.267	41.0%	2.05×
Top 30%	0.231	52.2%	1.74×
Top 40%	0.204	62.7%	1.57×
Top 50%	0.179	73.9%	1.48×

The top 20% of ranked customers capture 41.0% of all actual responders. Operationally, this means that the same response volume achieved through randomly targeting 40% of customers can instead be obtained by contacting only the top 20% predicted customers. This effectively halves the number of promotional contacts required while maintaining equivalent response volume.

The concentration of value within upper score deciles suggests that the model has learned meaningful behavioural signal rather than overfitting to random noise.

5.5 Feature Importance Analysis

Feature importance was evaluated using LightGBM split-based importance scores across the 179-iteration ensemble.

Most Influential Feature Groups

The strongest predictive signals originate from:

- 1. Historical shopping regularity
- 2. Spending intensity and variability
- 3. Brand and category engagement
- 4. Price sensitivity
- 5. Store-level response behaviour

Day-of-Week Behaviour

Day-of-week transaction count features collectively represent the most influential signal class. In particular, Wednesday purchase count (dow_3_count) ranked as the single most important feature with an importance score of 3,176. This suggests that the consistency and timing of customer shopping routines strongly influence promotional responsiveness.

Spending Behaviour Metrics

Several monetary variables also ranked highly:

Feature	Interpretation
amt_coefficient_variation	Spending consistency
amt_mean	Average transaction amount
amt_sum	Total spending
qty_mean	Average purchase quantity
amt_std	Spending variability

These features collectively capture purchasing intensity and long-term customer engagement.

Loyalty and Diversity Metrics

Brand and category diversity scores contributed additional predictive signal by measuring the breadth and concentration of customer shopping behaviour.

Customer tenure also contributed modestly, suggesting behavioural differences between newer and longer-established customers.

Price Sensitivity Indicators

Several price sensitivity variables ranked within the top predictive tier:

- Promotion value relative to customer average spending
- Promotion quantity relative to average purchase quantity
- Relative promotional attractiveness measures

These findings indicate that customer responsiveness depends partly on how attractive a promotion appears relative to historical spending habits.

Summary Interpretation

Overall, promotion responsiveness is driven primarily by:

- Temporal shopping patterns
- Spending intensity and consistency
- Brand affinity
- Price sensitivity
- Recent engagement behaviour

Store-level response rates, competition variables, and recency metrics provide additional contextual signal, while manufacturer-level variables and several binary interaction indicators contribute comparatively less predictive value.

5.6 Feature Effects – Partial Dependence Analysis

Feature importance scores indicate which variables influence the model most strongly, but they do not reveal the direction or shape of those relationships.

Partial dependence analysis was therefore conducted for two interpretable variables:

- `store_response_rate`
- `category_amt_sum`

Store Response Rate

The partial dependence curve for `store_response_rate` exhibits a broadly positive and approximately monotonic relationship. Customers associated with stores exhibiting historically strong promotional engagement tend to have elevated predicted response probabilities.

This suggests that store-level factors — including assortment, operational execution, and local demographics — contribute to an ambient promotional engagement effect beyond individual

customer behaviour.

Category Spending

The relationship between cumulative category spending (`category_amt_sum`) and predicted response probability is non-linear.

Response probability increases with category spending up to a moderate expenditure level before plateauing or slightly declining at very high spending levels.

This pattern suggests that:

- Light-to-moderate category purchasers remain promotionally responsive
- Extremely heavy purchasers may already purchase regardless of promotions
- Incremental promotional impact diminishes among highly saturated customers

These relationships align with intuitive business expectations and support the conclusion that the model has extracted meaningful behavioural signal rather than spurious correlations.

6. Challenges and Future Directions

Large-scale retail transaction modelling introduces several methodological and operational challenges.

6.1 Key Challenges

Data Sparsity and Irregular Behaviour

Many customers exhibit sparse or irregular purchase histories, which can destabilize behavioural metrics such as recency and frequency.

This issue was partially mitigated through:

- Aggregation strategies
- Bayesian smoothing
- Log transformations
- Robust behavioural normalization

High-Cardinality Categorical Variables

Customer identifiers, brands, and promotion IDs contain substantial cardinality.

Improper encoding could easily introduce overfitting risk. Bayesian-smoothed target encoding was therefore used to balance predictive signal retention against statistical stability.

Temporal Leakage Prevention

Strict prevention of temporal leakage required exact pre-promotion transaction filtering for every observation. Although computationally intensive, this step was essential to ensure realistic deployment validity.

Model Complexity and Overfitting

The final 71-feature model exhibits a moderate training-validation gap.

Attempts at feature removal did not improve validation performance, suggesting that regularization and calibration may be more effective future optimization strategies than aggressive dimensionality reduction.

6.2 Future Directions

Several future enhancements could further improve predictive performance and operational utility.

Temporal Modelling

Future work could incorporate richer temporal dynamics, including:

- Rolling behavioural trends
- Seasonal adjustment
- Event-driven purchase spikes
- Sequential behavioural modelling

Customer Segmentation

Cohort-based modelling or customer segmentation approaches may support more personalized promotional strategies.

Advanced Ensemble Methods

Potential modelling improvements include:

- Ensemble stacking
- Blended gradient boosting architectures
- Calibrated probability estimation
- Unsupervised customer pre-clustering

Probability Calibration

If probabilities are used directly for:

- Budget allocation
- ROI forecasting
- Expected-response estimation

then post-hoc calibration techniques such as isotonic regression or Platt scaling may improve reliability.

Production Monitoring

A deployment framework should continuously monitor:

- Post-campaign response rates
- Prediction drift
- Population stability
- Behavioural shifts over time

Such monitoring would enable early detection of concept drift as customer behaviour evolves across future campaign cycles.

7. Appendix – Probability Calibration

Predicted probabilities were evaluated for calibration quality — that is, whether predicted response probabilities correspond to observed response frequencies in held-out validation data.

A calibration curve was constructed by:

1. Dividing predictions into 10 equally spaced probability bins
2. Computing the mean predicted probability within each bin
3. Comparing predicted probabilities against observed response rates

The LightGBM model exhibits reasonable calibration across the low-to-mid probability range, where most predictions are concentrated given the 20.2% response base rate. At higher predicted probabilities, the model demonstrates a moderate tendency to underestimate actual response likelihood.

This behaviour is common among gradient boosting models trained on moderately imbalanced datasets because the optimization process encourages conservative assignment of high-confidence positive predictions.

For the primary use case of customer ranking and decile-based targeting, this calibration behaviour has limited operational impact because ranking quality — measured by AUC — is independent of absolute probability calibration.

However, if predicted probabilities are used directly for:

- Budget allocation
- Campaign ROI estimation

- Expected-response forecasting
- Financial planning

then post-hoc calibration methods such as isotonic regression or Platt scaling should be applied prior to deployment.

8. Key Takeaways

- The final LightGBM model achieved the strongest validation performance with an AUC of 0.7134.
- Cross-validation confirmed that model performance was stable and not dependent on a favourable data split.
- Historical shopping regularity emerged as the strongest predictor of promotional response.
- Spending behaviour, brand affinity, and price sensitivity also contributed substantial predictive signal.
- The model achieved meaningful business impact, including a 2.05× lift in responder capture within the top 20% of ranked customers.
- Leakage-free feature engineering and behavioural aggregation were critical to model validity.
- Future improvements may include temporal sequence modelling, calibrated probabilities, and production monitoring frameworks.

Overall, the project demonstrates that customer promotional responsiveness can be effectively modelled using historical transactional behaviour, enabling more efficient and data-driven campaign targeting strategies.